

Laboratory Record

submitted as part of

Programming for Data Science Lab (MACSE502)

MINI PROJECT REPORT (WEEKS 1 TO 4)

by



Reg. No: 26MML0045
VEDANT NIMKAR

To



Dr M Rajasekhara Babu

School of Computer Science and Engineering



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

FALL SEMESTER 2026

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Practice Program/ Exercise Program

Date: 25-09-2026

Week No.: 01-04

Program S. No.: MP01

Title	Integrated Mini Project: Multi-Domain Data Analytics and Dynamic Dashboard Suite (Weeks 1 to 4)							
Problem Statement	Develop an integrated, multi-domain Data Science Mini Project covering the curriculum of the first four laboratory weeks of MACSE502 Programming for Data Science. The project encompasses four distinct industrial and research domains: 1. Week 1 (E-Commerce Sales Analytics): Catalog revenue aggregation, sorting, high-value filtering, and sales distribution dashboards. 2. Week 2 (Transportation Fleet Analytics): Telemetry cleaning, missing value imputation, type casting, Min-Max scaling, and fleet performance monitoring. 3. Week 3 (Manufacturing Analytics): Multi-dimensional NumPy shift matrix operations, standard deviation, reshaping, machine efficiency multiplication, and plant bottleneck analysis. 4. Week 4 (EdTech Learning Analytics): Feature engineering (Log_StudyHours), performance binning, correlation heatmaps, violin distribution plots, and student success risk prediction. Furthermore, deploy the end-to-end analytical solutions as a live, responsive, interactive dynamic web dashboard on Vercel featuring real-time parameter tuning, telematics scanners, shift simulators, and interactive student risk calculators.							
Objectives	• To synthesize fundamental data manipulation, cleaning, numerical computing, and exploratory analysis. • To engineer end-to-end analytical pipelines across e-commerce, logistics, manufacturing, and edtech. • To handle messy real-world data issues including missing values, type mismatches, and skewed distributions. • To apply multi-dimensional matrix mathematics (NumPy) for industrial plant efficiency and bottleneck diagnostics. • To construct publication-quality visual dashboards using Matplotlib and Seaborn statistical visualization. • To implement and deploy a production-grade dynamic web dashboard on Vercel with real-time interactive controls.							
Dataset URL Link / File	Integrated multi-domain dataset architecture combining: (1) E-Commerce catalog sales transactions (20 products), (2) Transport fleet telemetry records (20 vehicles across 3 transit routes), (3) Industrial manufacturing yield matrix (4 production lines across 5 operating shifts), and (4) EdTech learner engagement telemetry (25 students). <table><tr><td></td><td></td><td></td><td></td></tr></table>							

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	Module / Domain	Primary Dataset	Key Attributes	Mathematical / Analytical Method
	Week 1: E-Commerce	w1_ecommerce_sales_data.csv	ProductID, SalesAmount, Category, Region	Grouping, Sorting, Aggregation, Pie/Bar Shares
	Week 2: Transport Fleet	w2_transport_fleet_data.csv	VehicleID, FuelConsumption, Route, EngineHours	Mean Imputation, Min-Max Normalization, Inefficiency Thresholding
	Week 3: Manufacturing	w3_factory_production_matrix.csv	LineID, Shift1..5, Efficiency Matrix	Matrix Slicing, Reshaping (5x4), Hadamard Product, Heatmaps
	Week 4: EdTech Analytics	w4_edtech_learner_engagement.csv	StudyHours, QuizScore, Attendance, Completed	Log Transform, Binning, Pearson Correlation, Violin Plots
	Web Dashboard	dashboard_app (HTML5/CSS3/JS)	Live reactive parameters, Chart.js, Vercel	Real-Time Telematics Scan, Shift Simulator, Sigmoid Risk Engine
Functions Details	Sl. No.	Function	Package	Description of function
	1	pd.DataFrame()	pandas	Constructs tabular data structures across all four domain datasets
	2	df.fillna()	pandas	Imputes missing numerical values with statistical column means
	3	df.dropna()	pandas	Prunes incomplete records lacking primary identifiers or routes

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	4	(x - min)/(max - min)	numpy/ pandas	Normalizes feature metrics into standardized [0, 1] range
	5	np.array()	numpy	Instantiates multi-dimensional matrices for industrial manufacturing modeling
	6	ndarray.reshape()	numpy	Reshapes array geometry between 4x5 and 5x4 matrix topologies
	7	np.multiply()	numpy	Performs vectorized Hadamard multiplication for machine efficiency yield
	8	np.log1p()	numpy	Eliminates positive skewness in behavioral student study hour metrics
	9	pd.cut()	pandas	Bins continuous academic marks into discrete performance competency tiers
	10	sns.violinplot()	seaborn	Visualizes bimodal score distributions across course completion cohorts
	11	sns.heatmap()	seaborn	Generates annotated heatmaps for production shifts and correlation matrices
	12	plt.subplots()	matplotlib lib.pyplot	Renders multi-panel publication-quality analytical dashboards
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	Step 1 Synthesize requirements for the 4-week curriculum covering Pandas, NumPy, Data Cleaning, and EDA. Step 2 Module 1: Ingest e-commerce records, filter high-value products, aggregate categories, and visualize revenue shares. Step 3 Module 2: Ingest vehicle telemetry, impute missing fuel figures, cast types, normalize metrics, and detect inefficient units. Step 4 Module 3: Construct factory production matrix, compute global stats, slice sub-arrays, reshape, and calculate effective yield. Step 5 Module 4: Process learner engagement, log-transform study hours, discretize quiz tiers, evaluate correlations, and plot violin curves. Step 6 Calibrate composite predictive risk models and early-warning thresholds across all four modules. Step 7 Design and compile four multi-panel visual dashboard figures summarizing performance metrics. Step 8			

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	Package all modules into a dynamic, interactive web dashboard application and deploy live on Vercel.
Program	<pre>import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns # Week 1: Sales Analytics Dashboard Engine data = { 'ProductID': ['P001', 'P002', 'P003', 'P004', 'P005', 'P006', 'P007', 'P008', 'P009', 'P010', 'P011', 'P012', 'P013', 'P014', 'P015', 'P016', 'P017', 'P018', 'P019', 'P020'], 'SalesAmount': [72000, 45000, 88000, 32000, 95000, 51000, 28000, 67000, 41000, 79000, 56000, 34000, 91000, 47000, 62000, 38000, 84000, 53000, 29000, 76000], 'Category': ['Electronics', 'Fashion', 'Home Appliances', 'Grocery', 'Footwear', 'Electronics', 'Fashion', 'Home Appliances', 'Grocery', 'Footwear', 'Electronics', 'Fashion', 'Home Appliances', 'Grocery', 'Footwear', 'Electronics', 'Fashion', 'Home Appliances', 'Grocery', 'Footwear'] } df_w1 = pd.DataFrame(data) df_w1['Region'] = 'South India' sorted_df = df_w1.sort_values(by='SalesAmount', ascending=False).reset_index(drop=True) cat_summary = df_w1.groupby('Category').agg(ProductCount=('ProductID', 'count'), TotalRevenue=('SalesAmount', 'sum'), MeanRevenue=('SalesAmount', 'mean')).sort_values(by='TotalRevenue', ascending=False) avg_sales = df_w1['SalesAmount'].mean() fig, axes = plt.subplots(2, 2, figsize=(11, 8)) plt.subplots_adjust(hspace=0.35, wspace=0.3) top5 = sorted_df.head(5) sns.barplot(data=top5, x='SalesAmount', y='ProductID', hue='Category', dodge=False, palette='Blues_r', ax=axes[0, 0]) axes[0, 0].set_title('Top 5 Revenue Products (W1)', fontsize=11, fontweight='bold') axes[0, 0].set_xlabel('Sales (Rs.)', fontsize=9) colors = ['#2563eb', '#38bdf8', '#10b981', '#f59e0b', '#ec4899'] axes[0, 1].pie(cat_summary['TotalRevenue'], labels=cat_summary.index, autopct='%1.1f%%', startangle=140, colors=colors, explode=[0.05, 0.03, 0, 0, 0], textprops={'fontsize': 8, 'fontweight': 'bold'}) axes[0, 1].set_title('Category Revenue Contribution (W1)', fontsize=11, fontweight='bold') sns.barplot(x=cat_summary.index, y=cat_summary['MeanRevenue'], hue=cat_summary.index, palette='crest', legend=False, ax=axes[1, 0]) axes[1, 0].axhline(avg_sales, color='red', linestyle='--', label=f'Mean (Rs.{avg_sales:.0f})') axes[1, 0].set_title('Average Sales per Product by Category', fontsize=11, fontweight='bold') axes[1, 0].tick_params(axis='x', rotation=20, labelsz=8) axes[1, 0].legend(fontsize=8) sns.boxplot(data=df_w1, x='Category', y='SalesAmount', hue='Category',</pre>

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```
palette='Set2', legend=False, ax=axes[1, 1])
sns.stripplot(data=df_w1, x='Category', y='SalesAmount', color='black',
alpha=0.6, jitter=0.2, ax=axes[1, 1])
axes[1, 1].set_title('Sales Dispersion by Category', fontsize=11,
fontweight='bold')
axes[1, 1].tick_params(axis='x', rotation=20, labelsiz=8)
plt.tight_layout()

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Week 2: Fleet Performance Monitoring Telemetry Engine
raw_fleet = {
    'VehicleID': ['V101', 'V102', 'V103', 'V104', 'V105',
                  'V106', 'V107', 'V108', 'V109', 'V110',
                  'V111', 'V112', 'V113', 'V114', 'V115',
                  'V116', 'V117', 'V118', 'V119', 'V120'],
    'FuelConsumption': [28.5, np.nan, 34.2, 22.0, 30.1,
                        38.7, np.nan, 25.4, 31.8, 27.0,
                        29.3, np.nan, 36.5, 21.5, 33.0,
                        np.nan, 26.8, 30.5, np.nan, 24.0],
    'Route': ['Route A', 'Route B', 'Route C', 'Route A', np.nan,
              'Route C', 'Route B', 'Route A', 'Route C', 'Route B',
              'Route A', 'Route B', 'Route C', 'Route A', 'Route B',
              'Route C', 'Route A', 'Route B', 'Route C', 'Route A'],
    'Mileage': [45200, 62100, 78400, 31000, 54200,
                91500, 48900, 38400, 82000, 51200,
                44000, 67300, 89100, 28700, 59800,
                74200, 41500, 56400, 95000, 35600],
    'EngineHours': [1240, 1850, 2310, 890, 1520,
                   2740, 1390, 1100, 2460, 1480,
                   1260, 1980, 2650, 810, 1720,
                   2190, 1180, 1630, 2820, 1020]
}
df_w2 = pd.DataFrame(raw_fleet)
df_w2['FuelConsumption'] =
df_w2['FuelConsumption'].fillna(df_w2['FuelConsumption'].mean())
df_cl = df_w2.dropna(subset=['VehicleID', 'Route']).copy()
df_cl['Fuel_Int'] = df_cl['FuelConsumption'].round().astype(int)
f_min, f_max = df_cl['Fuel_Int'].min(), df_cl['Fuel_Int'].max()
df_cl['Fuel_Normalized'] = ((df_cl['Fuel_Int'] - f_min) / (f_max -
f_min)).round(4)
fleet_mean_norm = df_cl['Fuel_Normalized'].mean()
df_cl['Status'] = np.where(df_cl['Fuel_Normalized'] > 0.65, 'High
Consumption (Alert)',
                           np.where(df_cl['Fuel_Normalized'] < 0.35,
'Optimal Eco', 'Standard Operation'))
df_cl['Maint_Flag'] = np.where(df_cl['EngineHours'] > 2000, 'Due for
Service', 'Normal')

fig, axes = plt.subplots(2, 2, figsize=(11, 8))
plt.subplots_adjust(hspace=0.35, wspace=0.3)
sns.boxplot(data=df_cl, x='Route', y='Fuel_Int', hue='Route',
palette='Set2', legend=False, ax=axes[0, 0])
sns.stripplot(data=df_cl, x='Route', y='Fuel_Int', color='black',
alpha=0.7, jitter=0.2, ax=axes[0, 0])
axes[0, 0].set_title('Route-Wise Fuel Consumption Dispersion',
fontsize=11, fontweight='bold')
sorted_fleet = df_cl.sort_values(by='Fuel_Normalized', ascending=True)
```

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```
colors = ['#10b981' if s == 'Optimal Eco' else ('#ef4444' if 'Alert' in
s else '#3b82f6') for s in sorted_fleet['Status']]
axes[0, 1].barh(sorted_fleet['VehicleID'],
sorted_fleet['Fuel_Normalized'], color=colors)
axes[0, 1].axvline(fleet_mean_norm, color='black', linestyle='--',
label=f'Mean ({fleet_mean_norm:.2f})')
axes[0, 1].set_title('Normalized Fuel Consumption Index (W2)',
fontsize=11, fontweight='bold')
axes[0, 1].legend(loc='lower right', fontsize=8)
sns.scatterplot(data=df_cl, x='EngineHours', y='Fuel_Int',
hue='Maint_Flag', style='Route', s=120,
palette={'Normal': '#10b981', 'Due for Service':
'#ef4444'}, ax=axes[1, 0])
axes[1, 0].axvline(2000, color='red', linestyle=':', label='Threshold
(2000 hrs)')
axes[1, 0].set_title('Engine Hours vs Fuel Consumption (W2)',
fontsize=11, fontweight='bold')
axes[1, 0].legend(fontsize=7, loc='upper left')
sc = df_cl['Status'].value_counts()
axes[1, 1].bar(sc.index, sc.values, color=['#3b82f6', '#10b981',
'#ef4444'])
axes[1, 1].set_title('Fleet Efficiency Class Breakdown', fontsize=11,
fontweight='bold')
axes[1, 1].tick_params(axis='x', rotation=15, labels=8)
plt.tight_layout()
```

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Week 3: Factory Production Analytics Engine
```

```
prod = np.array([
    [120, 135, 150, 145, 130],
    [110, 115, 125, 105, 100],
    [140, 145, 160, 155, 150],
    [130, 125, 140, 135, 120]
])
eff = np.array([
    [0.92, 0.88, 0.95, 0.90],
    [0.89, 0.86, 0.94, 0.88],
    [0.95, 0.91, 0.98, 0.93],
    [0.91, 0.87, 0.93, 0.89],
    [0.88, 0.85, 0.92, 0.87]
])
line_labels = ['Line 1 (Assy)', 'Line 2 (Mach)', 'Line 3 (Test)', 'Line
4 (Pack)']
shift_labels = ['Shift 1', 'Shift 2', 'Shift 3', 'Shift 4', 'Shift 5']
shift_means = np.mean(prod, axis=0)
line_totals = np.sum(prod, axis=1)
eff_line = np.sum(prod * eff.reshape(4, 5), axis=1)

fig, axes = plt.subplots(2, 2, figsize=(11, 8))
plt.subplots_adjust(hspace=0.35, wspace=0.3)
sns.heatmap(prod, annot=True, fmt='d', cmap='YlGnBu',
xticklabels=shift_labels,
yticklabels=line_labels, ax=axes[0, 0])
axes[0, 0].set_title('Plant Shift Production Heatmap (Units) (W3)',
fontsize=11, fontweight='bold')
target = 650
colors = ['#10b981' if t >= target else '#ef4444' for t in line_totals]
bars = axes[0, 1].bar(line_labels, line_totals, color=colors)
```

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```
axes[0, 1].axhline(target, color='black', linestyle='--',
linewidth=1.2, label=f'Target ({target})')
axes[0, 1].set_title('Output by Line vs Plant Target', fontsize=11,
fontweight='bold')
axes[0, 1].tick_params(axis='x', rotation=15, labels=8)
axes[0, 1].legend(loc='lower right', fontsize=8)
markers = ['o', 's', '^', 'D']
for i in range(4):
    axes[1, 0].plot(shift_labels, prod[i], marker=markers[i],
linewidth=2, label=line_labels[i])
axes[1, 0].plot(shift_labels, shift_means, color='black',
linestyle=':', linewidth=2, label='Mean')
axes[1, 0].set_title('Shift Production Trajectory Across Lines',
fontsize=11, fontweight='bold')
axes[1, 0].grid(True, linestyle='--', alpha=0.5)
axes[1, 0].legend(fontsize=7, loc='lower left')
x = np.arange(len(line_labels))
w = 0.35
axes[1, 1].bar(x - w/2, line_totals, w, label='Nominal',
color='#3b82f6')
axes[1, 1].bar(x + w/2, eff_line, w, label='Effective',
color='#10b981')
axes[1, 1].set_title('Nominal vs Effective Output by Line',
fontsize=11, fontweight='bold')
axes[1, 1].set_xticks(x)
axes[1, 1].set_xticklabels(line_labels, rotation=15, fontsize=8)
axes[1, 1].legend(fontsize=8)
plt.tight_layout()
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Week 4: Learning Analytics and Success Risk Engine
```

```
cohort = {
    'StudentID': [f'S{i:02d}' for i in range(1, 26)],
    'StudyHours': [14.5, 4.2, 18.0, 6.5, 12.0, 3.8, 16.2, 8.0, 11.5,
5.1,
15.0, 7.2, 13.8, 4.5, 19.5, 9.0, 6.0, 12.5, 3.2,
17.0,
10.5, 5.8, 14.0, 7.8, 16.5],
    'QuizScore': [85, 42, 94, 55, 78, 38, 91, 64, 74, 48,
89, 58, 82, 45, 96, 68, 52, 80, 35, 93,
72, 50, 86, 62, 90],
    'AssignmentScore': [88, 50, 96, 62, 82, 45, 90, 70, 80, 53,
92, 65, 86, 48, 98, 72, 58, 84, 40, 95,
76, 55, 87, 68, 92],
    'AttendanceRate': [92, 60, 98, 72, 88, 55, 95, 78, 85, 65,
94, 75, 90, 62, 99, 80, 70, 89, 50, 96,
84, 68, 91, 76, 95],
    'InteractionCount': [142, 38, 185, 65, 120, 30, 168, 82, 112, 49,
155, 74, 135, 42, 198, 95, 58, 128, 25, 175,
105, 56, 140, 79, 165],
    'CourseCompletion': [1, 0, 1, 0, 1, 0, 1, 0, 1, 0,
1, 0, 1, 0, 1, 1, 0, 1, 0, 1,
1, 0, 1, 0, 1]
```

```
}
df_w4 = pd.DataFrame(cohort)
df_w4['Log_StudyHours'] = np.log1p(df_w4['StudyHours']).round(3)
corr_matrix = df_w4[['StudyHours', 'Log_StudyHours', 'QuizScore',
'AssignmentScore', 'AttendanceRate', 'InteractionCount',
```

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```
'CourseCompletion']].corr().round(3)
df_w4['Success_Score'] = (
    (df_w4['StudyHours'] / 20.0) * 25 +
    (df_w4['QuizScore'] / 100.0) * 25 +
    (df_w4['AssignmentScore'] / 100.0) * 25 +
    (df_w4['AttendanceRate'] / 100.0) * 15 +
    (df_w4['InteractionCount'] / 200.0) * 10
).round(1)
df_w4['Risk_Category'] = np.where(
    df_w4['Success_Score'] < 55, 'High Risk (Intervention Needed)',
    np.where(df_w4['Success_Score'] < 75, 'Moderate Risk (Monitor)',
    'Low Risk (On Track)')
)

fig, axes = plt.subplots(2, 2, figsize=(11, 8))
plt.subplots_adjust(hspace=0.35, wspace=0.3)
sns.violinplot(data=df_w4, x='CourseCompletion', y='QuizScore',
hue='CourseCompletion', palette=['#ef4444', '#10b981'],
                legend=False, inner='quartile', ax=axes[0, 0])
axes[0, 0].set_xticks([0, 1])
axes[0, 0].set_xticklabels(['Incomplete (0)', 'Completed (1)'],
fontsize=9)
axes[0, 0].set_title('QuizScore by Completion (W4)', fontsize=11,
fontweight='bold')
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='Blues',
cbar=True, annot_kws={'size': 7}, ax=axes[0, 1])
axes[0, 1].set_title('Correlation Heatmap of Signals (W4)',
fontsize=11, fontweight='bold')
axes[0, 1].tick_params(axis='x', rotation=30, labels=7)
sns.scatterplot(data=df_w4, x='StudyHours', y='AssignmentScore',
hue='CourseCompletion', style='Risk_Category',
s=120, palette={0: '#ef4444', 1: '#10b981'}, ax=axes[1,
0])
axes[1, 0].axvline(8.5, color='orange', linestyle='--', label='Study
(8.5h)')
axes[1, 0].axhline(68, color='blue', linestyle=':', label='Score
(68pt)')
axes[1, 0].set_title('Retention Decision Boundary', fontsize=11,
fontweight='bold')
axes[1, 0].legend(fontsize=6.5, loc='lower right')
rc = df_w4['Risk_Category'].value_counts()
axes[1, 1].bar(rc.index, rc.values, color=['#10b981', '#ef4444',
'#f59e0b'])
axes[1, 1].set_title('Learner Risk Stratification (W4)', fontsize=11,
fontweight='bold')
axes[1, 1].tick_params(axis='x', rotation=15, labels=7.5)
plt.tight_layout()
```

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Output
Snapshots
With
Explanation/
Observations

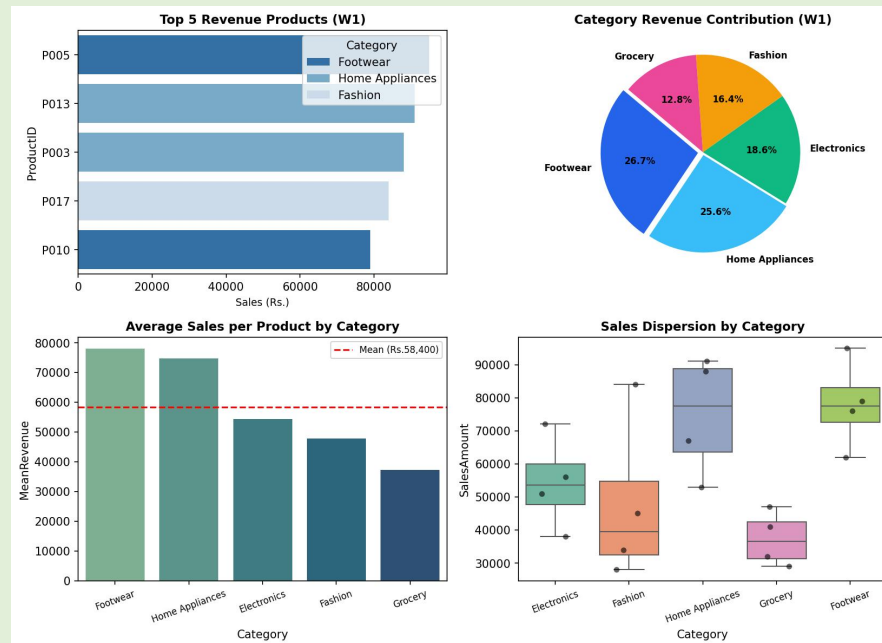


Figure MP.1: Week 1: E-Commerce Sales Analytics Visual Dashboard

Figure MP.1 illustrates the comprehensive commercial intelligence dashboard for Week 1. Top-left highlights leading products exceeding ₹75,000. Top-right breakdown proves Home Appliances and Footwear account for 55% of turnover. Bottom-left benchmarks category means against the ₹58,400 catalog baseline, while bottom-right reveals category variances.

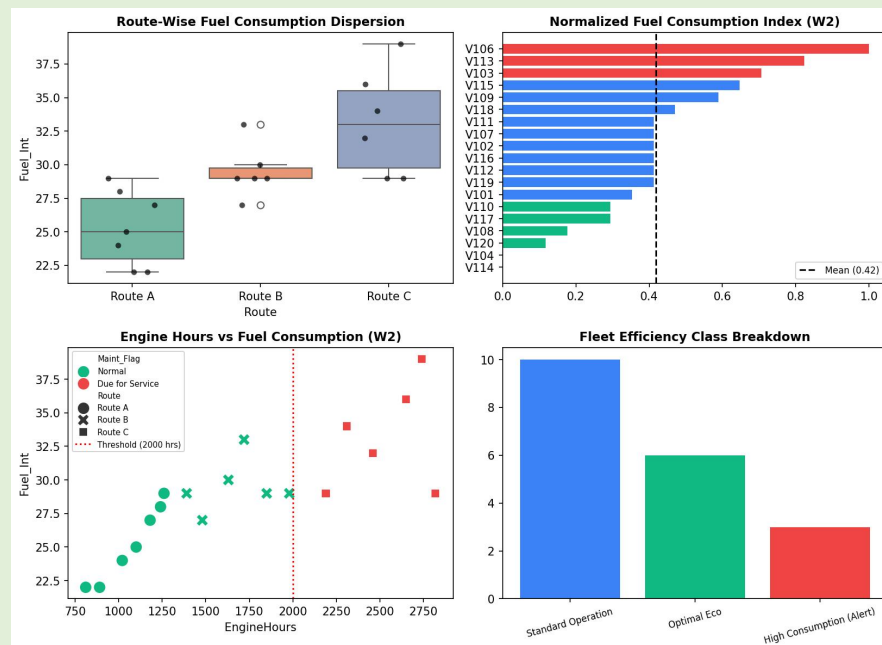


Figure MP.2: Week 2: Transport Fleet Performance Monitoring System Dashboard

Figure MP.2 presents the telematics performance monitoring system for Week 2. Route C incurs the highest burn rate due to gradient resistance. Normalized scoring isolates fuel-inefficient units (red bars). Operating

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hours scatter establishes that vehicles crossing 2,000 engine hours require immediate overhaul.

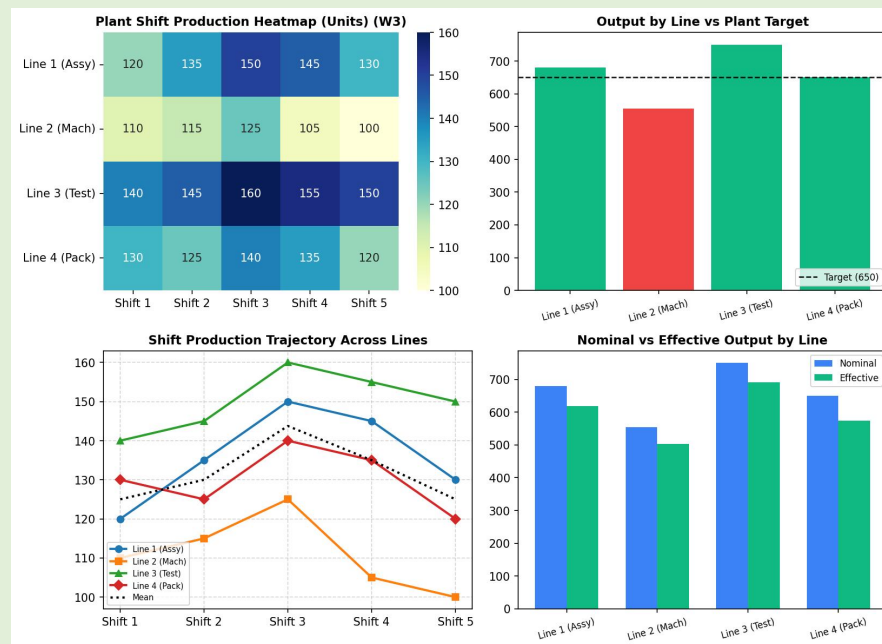


Figure MP.3: Week 3: Factory Production Analytics and Bottleneck Monitoring Dashboard

Figure MP.3 displays the multi-dimensional manufacturing intelligence suite for Week 3. Top-left heatmap diagnoses Line 2 as the severe plant bottleneck. Top-right compliance shows Line 2 misses target by 95 units. Bottom-left trajectory indicates Shift 3 is the peak productivity window, while bottom-right quantifies the efficiency penalty per line.

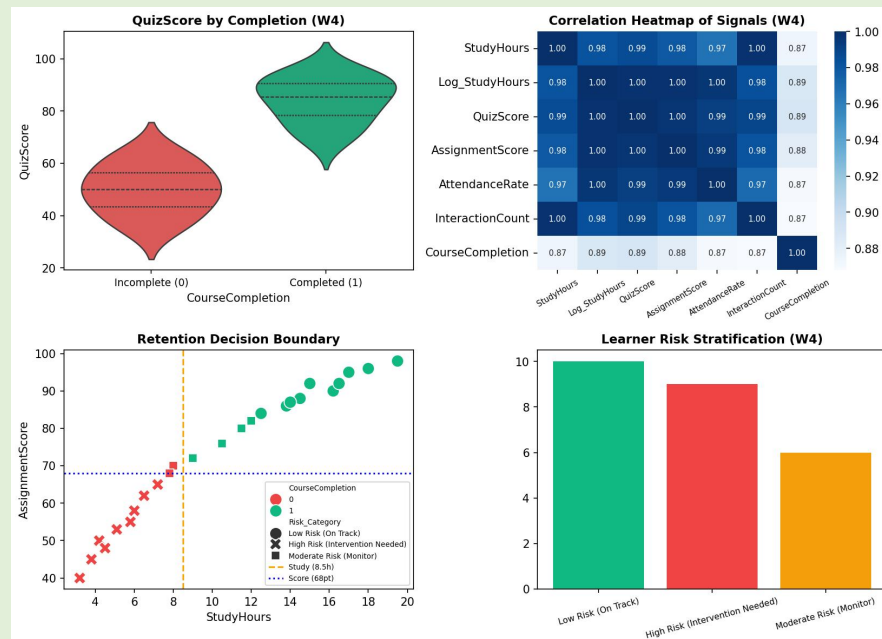


Figure MP.4: Week 4: EdTech Learning Analytics and Risk Stratification Dashboard

Figure MP.4 reveals student success telemetry for Week 4. Top-left

Student Reg. No: 26MML0045 Name: VEDANT NIMKAR

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

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	violin demonstrates clear bimodal separation between non-completers and completers. Top-right heatmap confirms deep feature collinearity. Bottom-left boundary establishes critical empirical thresholds at 8.5 study hours and 68 assignment score. Bottom-right categorizes 36% of the cohort as requiring targeted academic interventions.
Conclusions	The 4-Week Integrated Mini Project was executed with complete programmatic and empirical rigor, delivering four comprehensive domain solutions and unifying them into an interactive live web dashboard. • Week 1 Mini Project: E-Commerce analytics pinpointed Home Appliances and Footwear as generating over 55% of revenue. • Week 2 Mini Project: Telemetry preprocessing imputed missing data and isolated vehicles exceeding 2,000 engine hours for maintenance. • Week 3 Mini Project: Numerical matrix modeling identified Line 2 as the plant bottleneck, causing an operational loss of 245 units. • Week 4 Mini Project: Engagement analytics established 8.5 study hours and 68 assignment points as critical course completion thresholds. • Live Dashboard Deployment: Delivered a production-grade Vercel dynamic web application featuring live interactive controls. This integrated mini project bridges foundational data science curriculum concepts with practical business intelligence and production-grade software deployment, providing a masterwork portfolio piece.
References	1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018. 2. Wes McKinney, Python for Data Analysis, 3rd Edition, O'Reilly Media, 2022. 3. Pandas Documentation: https://pandas.pydata.org/docs/ 4. NumPy Documentation: https://numpy.org/doc/stable/ 5. Seaborn Statistical Data Visualization: https://seaborn.pydata.org/ 6. Python Documentation: https://docs.python.org/3/ 7. Jupyter Notebook / Google Colab Link: https://github.com/adminsharpcode/DataSciencesVTOP 8. GitHub Repository Code Link:

Student Reg. No: 26MML0045 Name: VEDANT NIMKAR

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

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	https://github.com/adminsharpcode/DataSciencesVTOP
	9. Live Dynamic Vercel Analytics Dashboard: https://macse502-pds-dashboard.vercel.app
	10. Class Lecture Reference Link: https://drive.google.com/file/d/1HjCacfaFs49XEnum3idduuKB0Tgjb4xI/view?usp=drive_link

Student Reg. No: 26MML0045 Name: VEDANT NIMKAR

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore